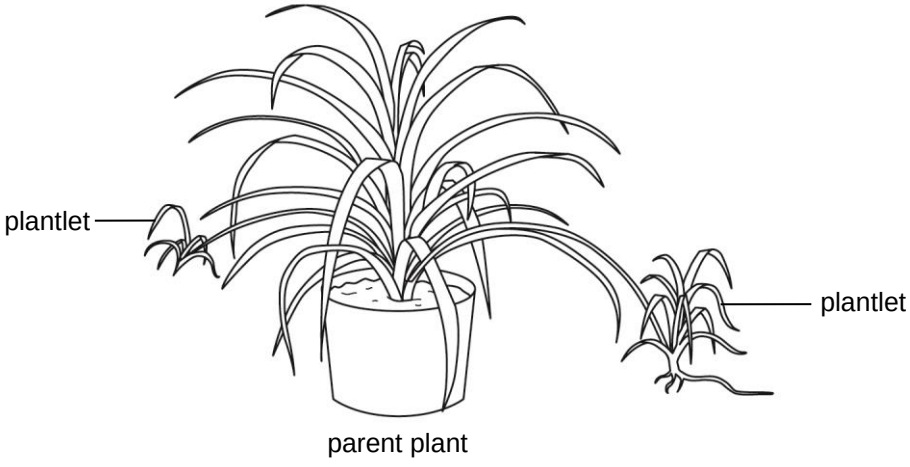
Multiple-choice questions

|!|BCEMB0130000002|!| ★

The diagram below shows a spider plant with two plantlets. Each plantlet can be separated from the parent plant and grown in soil.



This is an example of asexual reproduction because

A. the parent plant does not have flowers.
B. one parent plant can produce two daughter plants.
C. it does not involve the fusion of gametes.
D. the parent plants and the plantlets look the same.

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000004|!| ★

If we cut a potato tuber randomly into pieces of irregular sizes and shapes and grow them under favorable conditions, only some of the pieces can give rise to new potato plants. Which of the following explanations is correct?

A. The small sized pieces do not have enough food to support growth.
B. Only the pieces containing buds can grow.
C. The tissue cells of some pieces are damaged.
D. Some pieces may be infected by bacteria or fungi.

□

##B##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMC0130000005|!|
Which of the following flowers is most likely to be visited by the greater number of insects?

Flower	Size	Colour	Smell
(1)	small	white	scented
(2)	small	brightly-coloured	not scented
(3)	large	brightly-coloured	scented
(4)	large	brightly-coloured	not scented

A. (1)
B. (2)
C. (3)
D. (4)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0130000006|!|
Which of the following promotes seed dispersal?
A. butterflies visiting a flower
B. anthers shaking by wind
C. radicles breaking out of the seed coat
D. peapods popping open

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0130000007|!|

The following are the events that occur after pollination.

(1) The male gamete is released into the ovule to fertilize the ovum.
(2) The pollen tube enters the ovule through the micropyle.
(3) Each pollen grain develops a pollen tube.
(4) The pollen tube grows along the style towards the ovary.

Which of the following shows the correct sequence of the events?

A. (2) → (3) → (4) → (1)
B. (2) → (4) → (1) → (3)
C. (3) → (4) → (2) → (1)
D. (3) → (2) → (4) → (1)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0130000008|!| ★

Which of the following processes will most likely produce apples of the same variety and quality?

A. growing new apple trees from cuttings
B. growing new apple trees from seeds that come from the same parent plant
C. growing new apple trees from seeds that come from the same apple
D. growing new apple trees under the same environmental conditions

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0130000009|!| ★

Which of the following is a characteristic of a plant with insect-pollinated flowers?

A. They produce very large numbers of pollen grains.
B. They have rough and sticky pollen grains.
C. All their flowers have insect guides and nectaries.
D. They produce fleshy and sweet fruits.

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##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0130000010|!| ★

The photographs below show four different plant structures.

P ***Q*** ***R*** ***S***



Which of the following structures can give rise to offspring that are genetically identical to their parents naturally?

A. P and S only
B. Q and R only
C. P, Q and S only
D. Q, R and S only

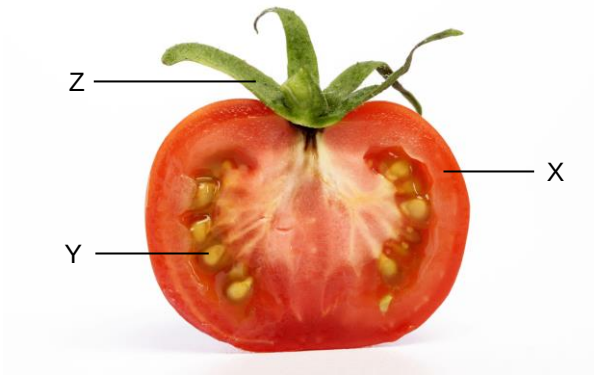
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000012|!| ★

The following photo shows the vertical section of a tomato.



Which of the following correctly identifies the floral parts from which structures X, Y and Z have developed?

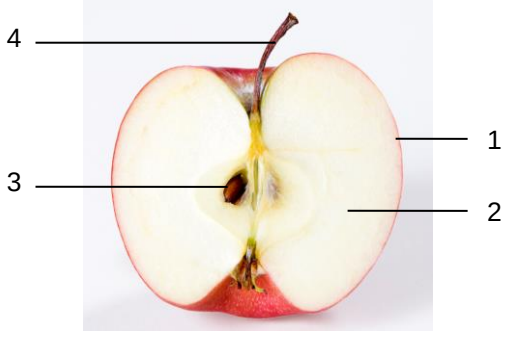
	X	***Y***	***Z***
A.	ovary	ovule	style
B.	ovule	ovary	style
C.	ovary	ovule	sepal
D.	ovule	ovary	sepal

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0130000013|!| ★

The following photo shows the longitudinal section of an apple.



Which of the following parts of the apple has different genetic makeup as the other parts?

A. 1

B. 2

C. 3

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D. 4

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000014|!|

Which of the following organisms can carry out asexual reproduction?

(1) *Amoeba*
(2) potato
(3) onion

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000015|!|

Which of the following statements about an onion bulb are correct?

(1) It is an underground stem.
(2) Food is stored in its fleshy scale leaves.
(3) It develops into a new plant from its buds.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0130000016|!| ★

Which of the following comparisons between binary fission and vegetative propagation is/are correct?

	Binary fission	***Vegetative propagation***
(1)	produces offspring asexually	produces offspring sexually
(2)	found in prokaryotes	found in flowering plants
(3)	does not involve mitosis	involves mitosis

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0130000017|!|

The following photograph shows some storage organs of plants.



The storage organs shown are

A. rhizomes.
B. corms.
C. stem tubers.
D. bulbs.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000018|!|

The carpel of a flower is composed of

(1) stigma.
(2) anther.
(3) ovary.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0130000019|!|

Which of the following statements about the structure of stamens is correct?

A. An anther is supported by a style.
B. Pollen grains are haploid.
C. Male gametes are made in the pollen sacs.
D. An anther contains two pollen sacs.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0130000020|!|

In which of the following floral parts are male gametes made?

A. style
B. anther
C. corolla
D. ovary

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0130000022|!|

Which of the following combinations ***incorrectly*** matches the floral part with the seed structure developed from the floral parts after fertilization?

	Floral part	*Seed structure*
A.	integument	seed coat
B.	zygote	embryo
C.	ovule	seed
D.	micropyle	hilum

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0130000023|!|

Which of the following plants relies on wind for seed dispersal?

A. dandelion
B. coconut
C. pea
D. tomato

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0130000025|!|

The following figure shows a fruit.



How is this fruit dispersed?

A. by wind
B. by animals
C. by water
D. by ‘explosion’ of the fruit wall

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000026|!|

Which of the following statements about vegetative propagation is ***incorrect***?

A. It can be induced artificially.
B. It decreases the chance of disease transmission from the parents to the offspring.
C. It produces offspring that are genetically identical to the parent.
D. It involves mitosis.

##B##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000027|!|

Which of the following comparisons between asexual and sexual reproductions is ***incorrect***?

	Asexual reproduction	***Sexual reproduction***
A.	Offspring are genetically identical to parents.	Offspring are genetically different from parents.
B.	No gametes are formed.	Gametes are formed.
C.	Fertilization does not take place.	Fertilization takes place.
D.	Only mitotic cell division is involved.	Only meiotic cell division is involved.

##D##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000028|!|

Which of the following is/are classified as asexual reproduction?

(1) binary fission in bacteria
(2) seed production in flowering plants
(3) vegetative propagation in flowering plants

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concept	1

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|!|BCEMB0130000030|!| ★

Which of the following comparisons between asexual and sexual reproductions is ***incorrect***?

	Asexual reproduction	***Sexual reproduction***
A.	involves one parent	usually involves two parents
B.	occurs in prokaryotes only	occurs in eukaryotes only
C.	does not involve gametes	involves gametes
D.	produces genetically identical offspring	produces genetically different offspring

##B##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000031|!|

The main advantage of sexual reproduction over asexual reproduction is that sexual reproduction provides

A. a higher chance of fertilization.
B. a higher chance of production of offspring with desirable characteristics.
C. a higher reproduction rate.
D. greater genetic variations in offspring.

##D##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000032|!|

Which of the following statements about different storage organs of plants is ***incorrect***?

A. An onion bulb stores food in fleshy scale leaves.
B. A ginger rhizome grows vertically underground.
C. A potato tuber stores food as starch.
D. A *Crocus* corm has no fleshy scale leaves.

##B##

Skill assessed	Mark
Understanding basic concept	1

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|!|BCEMC0130000033|!|

Which of the following statements about the vegetative propagation is/are correct?

(1) It may have a higher chance of transmitting disease from parent to offspring.
(2) It produces offspring which are genetically identical to the parent.
(3) It may lead to overcrowding.

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000035|!|

Which of the following structures is/are the component(s) of the stamen of a flower?

(1) anther
(2) filament
(3) stigma

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concept	1

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|!|BCEMC0130000036|!|

Which of the following structures is/are the component(s) of the carpel of a flower?

(1) ovary
(2) filament
(3) style

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000037|!|

Which of the following statements about insect-pollinated flowers is ***incorrect***?

A. The anthers are inside the flowers.
B. The petals are brightly coloured.
C. The stigma is broad and sticky.
D. The pollen grains are light and smooth.

##D##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMB0130000038|!| ★

Which of the following statements about self-pollination and cross-pollination is/are correct?

(1) No external agents are needed in self-pollination.
(2) Both self-pollination and cross-pollination result in genetic variations.
(3) External agents are needed in cross-pollination.

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A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000039|!|

Which of the following is ***not*** an advantage of vegetative propagation?

A. The desirable traits can be retained in the offspring.
B. The offspring are more likely to adapt to environmental changes.
C. The parent plant can reproduce without the involvement of another plant.
D. A large amount of food reserve is available for the growth of new plants.

##B##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000040|!|

Which of the following statements about potato tubers are correct?

(1) They are the swollen ends of underground roots.
(2) They are underground organs that store starch.
(3) They contain many ‘eyes’ which will give rise to new shoots in the next growing season.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concept	1

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|!|BCEMC0130000041|!|

Which of the following comparisons between insect-pollinated and wind-pollinated flowers is ***incorrect***?

		Insect-pollinated flowers	***Wind-pollinated flowers***
A.	Anthers	inside the petals	outside the petals
B.	Stigmas	sticky	feathery
C.	Pollen grains	lighter	heavier
D.	Petals	brightly coloured	dull in colour

##C##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMB0130000042|!| ★

The function(s) of fleshy fruit walls in fruits is/are

(1) to help seed dispersal by water.
(2) to provide food for seed germination.
(3) to help seed dispersal by animals.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000043|!|

Which of the following are the significances of seed dispersal?

(1) to reduce competition between offspring and the parents
(2) to avoid overcrowding of the plants
(3) to allow plants to colonize new areas

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A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##D##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMC0130000044|!|

A typical flower contains the following structures.

(1) ovary wall
(2) ovules
(3) petals
(4) sepals
(5) stamens

Which of the following shows the correct sequence of the locations of these structures starting from the outermost part of the typical flower?

A. (3) → (4) → (1) → (2) → (5)
B. (3) → (4) → (5) → (1) → (2)
C. (4) → (3) → (1) → (5) → (2)
D. (4) → (3) → (5) → (1) → (2)

□

##D##

Skill assessed	Mark
Understanding basic concept	1

|!|BCEMB0130000045|!| ★

The following diagram shows a structure of a dandelion.



Which of the following statements about this structure is/are correct?

(1) This structure can be carried by wind.
(2) This structure is usually produced in a large number to increase the chance of seed development into new plants.
(3) This structure can cling on animals.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000048|!|
Which of the following does a wind-pollinated flower generally have?

(1) styles with feathery stigmas
(2) smooth and light pollen grains
(3) stigmas well above the anthers

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0130000049|!|

Which of the following are the adaptive features of an insect-pollinated flower?

(1) Nectaries are present at the base of the corolla.
(2) Large amounts of smooth pollen grains are produced by the anthers.
(3) The filaments are short, keeping the anthers within the flower.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0130000051|!|

Which of the following about the fate of different parts of a flower after fertilization is correct?

	Before fertilisation	***After fertilisation***
A.	petal	fruit wall
B.	ovule	embryo
C.	integument	seed coat
D.	ovum	seed

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0130000052|!| ★

An experiment was carried out using flower buds growing on three separate plants of the same species.

Plant P: The anthers are removed and the flower buds are left open.
Plant Q: The anthers are removed and the flower buds are covered by plastic bags.
Plant R: The anthers are left intact and the flower buds are covered by plastic bags.

Later flowers develop on all the plants, but only those on plant P produce seeds.

This result shows that in this species,

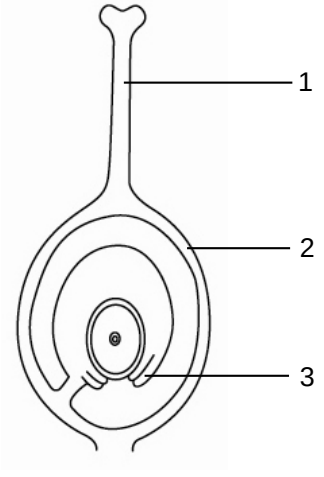
A. only insect-pollination can take place.
B. only wind-pollination can take place.
C. only self-pollination can take place.
D. only cross-pollination can take place.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0130000053|!|

The diagram below shows the carpel of a flower.



Which of the following correctly shows the fate of each part after fertilization?

	1	***2***	***3***
A.	petiole	falls off	seed coat
B.	falls off	fruit wall	cotyledon
C.	falls off	fruit wall	seed coat
D.	petiole	falls off	cotyledon

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##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000054|!|

Which of the following statements about vegetative propagation in flowering plants is / are correct?

(1) No pollination is required.
(2) It occurs in flowers.
(3) It involves vegetative parts (e.g. stems and roots).

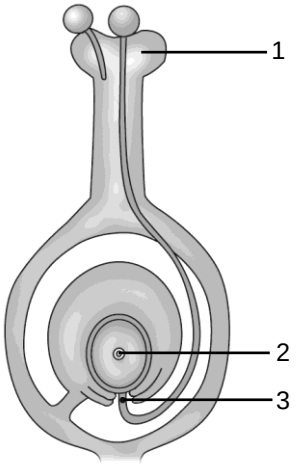
A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0130000059|!|

The following diagram shows some parts of a flower and two germinating pollen grains.



Which of the following combinations about the chromosome number of structures 1 to 3 is correct?

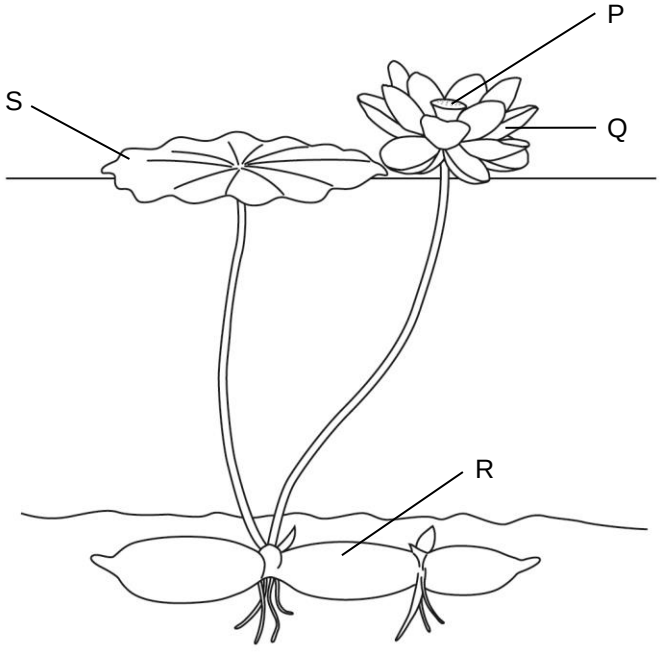
	Structure 1	*Structure 2*	*Structure 3*
A.	diploid	haploid	diploid
B.	haploid	diploid	diploid
C.	diploid	haploid	haploid
D.	haploid	diploid	haploid

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000060|!| ★

Directions: The following three questions refer to the diagram below, which shows the structure of a lotus plant.



Which of the following labelled structures is/are involved in the reproduction of this lotus plant?

A. P only
B. P and Q only
C. P and R only
D. P, Q, R and S

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##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000061|!| ★

What kind of storage organ is R?

A. stem tuber
B. bulb
C. corm
D. rhizome

□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000062|!| ★

People usually grow lotus plants by vegetative propagation because

(1) this method can produce offspring of better quality.
(2) it takes a shorter time to reproduce.
(3) less space is required.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

□

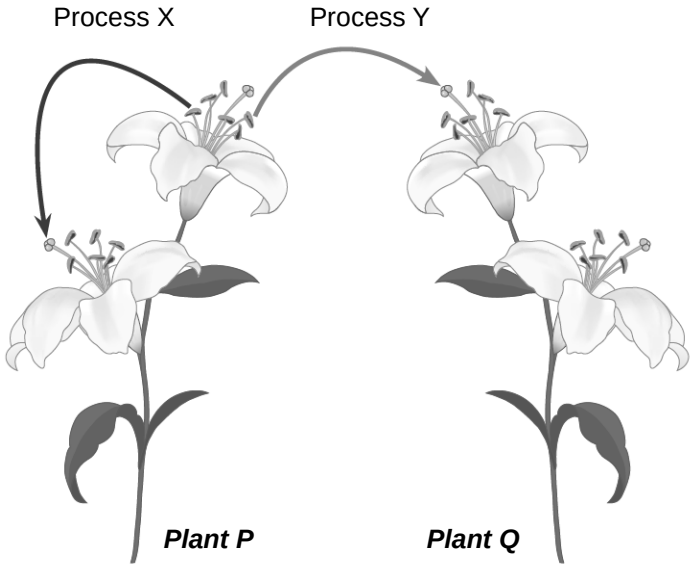
##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0130000063|!| ★

Directions: The following two questions refer to the diagram below, which shows two processes occurring in two flowering plants of the same species.



Which of the following combinations is correct?

	Process X	***Process Y***
A.	insect pollination	wind pollination
B.	cross-pollination	cross-pollination
C.	self-pollination	cross-pollination
D.	cross-pollination	self-pollination

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000064|!| ★

A series of events occur leading to fertilization after process Y has completed. Which of the following will ***not*** occur?

A. When a pollen grain reaches the stigma, it carries out meiotic cell division to develop a pollen tube.
B. The pollen tube carries male gametes.
C. The pollen tube grows along the style towards the ovary.
D. The male gamete is released to fuse with the female gamete in the ovule.

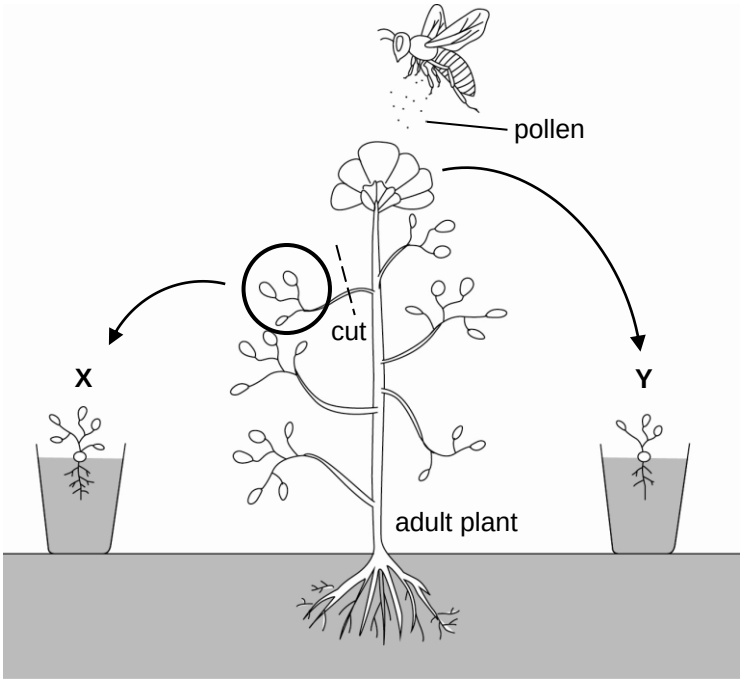
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##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000065|!| ★

Directions: The following two questions refer to the following diagram which shows the two methods that can be used to grow new plants from an adult plant.



Which of the following are the characteristics of the flower of the adult plant?

(1) Anthers are inside the flower.
(2) Smooth pollen grains are produced in large number by the flower.
(3) Nectaries are present inside the flower.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000066|!| ★

Which of the following comparisons between the methods of producing plants X and Y is correct?

		Method of producing plant X	***Method of producing plant Y***
A.	Genetic variations in offspring	more	less
B.	Chromosome number of the cells of offspring	haploid	diploid
C.	Speed of reproduction	slower	faster
D.	Chance of disease transmission to offspring	higher	lower

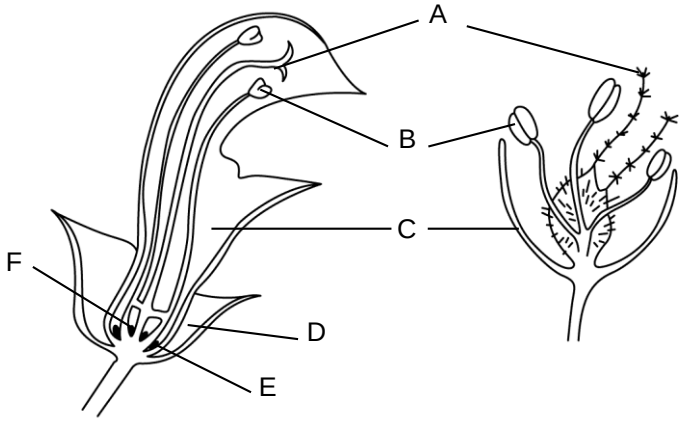
□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0130000067|!| ★

Directions: The following two questions refer to the diagrams below which show two kinds of flowers.



Flower X **Flower Y**

Which of the following structures will develop into seeds?

A. structure B
B. structure D
C. structure E
D. structure F

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000068|!| ★

Which of the following comparisons between Flower X and Flower Y is ***incorrect***?

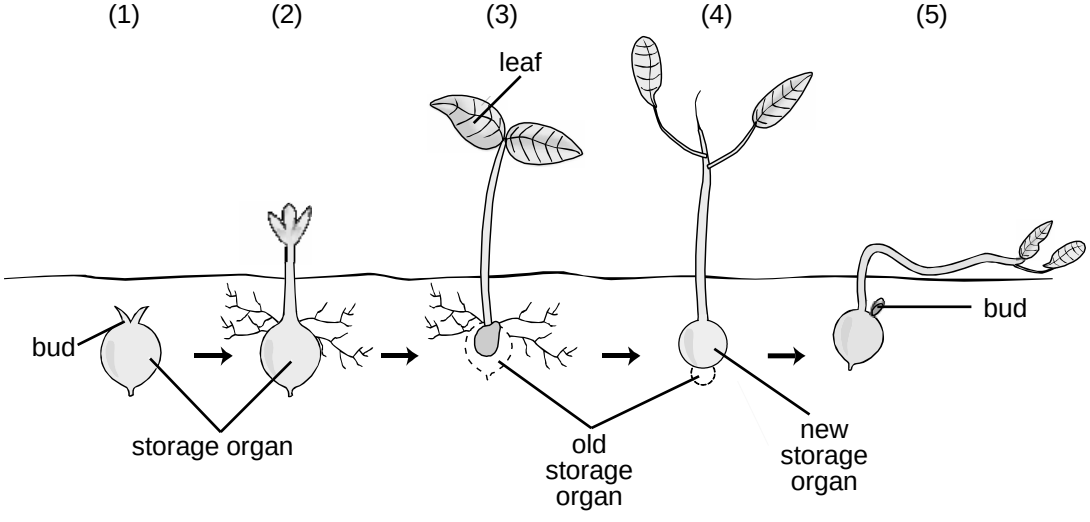
	Flower X	***Flower Y***
A.	with brightly coloured structure C	with dull coloured structure C
B.	with sticky structure A	with feathery structure A
C.	nectaries are abesent	nectaries are present
D.	structure B produces fewer gametes	structure B produces more gametes

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000069|!|

Directions: The following two questions refer to the diagram below which shows the stages of the development of an underground storage organ of a flowering plant.



Which of the following statements about the above stages are correct?

(1) In stage (2), shoots and roots grow using the food reserves in the storage organ.
(2) In stage (3), food is supplied by the storage organs and photosynthesis for the growth.
(3) In stage (4), the rate of photosynthesis exceeds that of respiration. Excess food is stored in the old storage organ.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000070|!|
Which of the following is/are the benefits of the development of underground storage organs to the plant?

(1) absorb more water
(2) reproduce asexually
(3) survive during cold winter

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

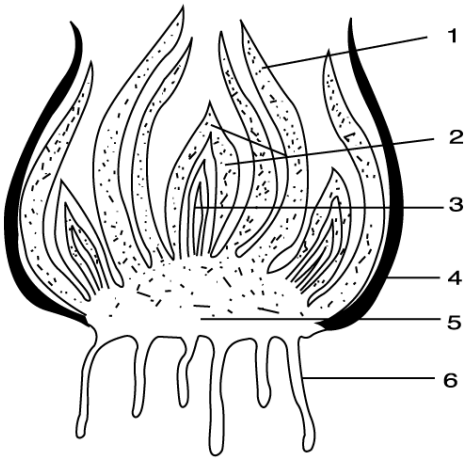
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0130000071|!| ★

Directions: The following three questions refer to the diagram below which shows a longitudinal section of an onion.



Which of the following structures will develop into a flower?

A. 1

B. 3

C. 4

D. 5

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000072|!| ★

Which of the following structures is a fleshy scale leaf?

A. 1

B. 3

C. 4

D. 5

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000073|!| ★

The function(s) of structure 6 is/are

(1) for anchorage.
(2) to absorb water and minerals.
(3) to form new storage organs.

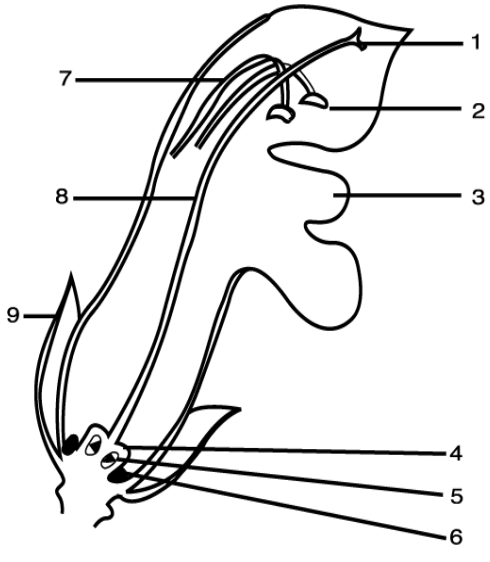
A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0130000074|!| ★

Directions: The following three questions refer to the diagram below which shows a longitudinal section of a flower.



In which of the following structures does fertilization occur?

A. 1
B. 2
C. 5
D. 8

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000075|!| ★

The structures that serve to attract insects for pollination include

A. 1 and 2.
B. 2 and 7.
C. 3 and 6.
D. 3 and 9.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000076|!| ★

Self-pollination in this flower may be avoided because

A. it is a unisexual flower.
B. the ovary is below the anthers.
C. the anthers are inside the flowers.
D. the stigma protrudes above the anthers.

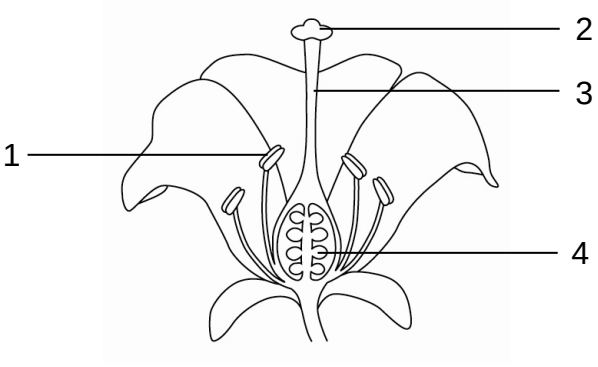
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0130000077|!|

Directions: The following two questions refer to the diagram below which shows a longitudinal section of a flower.



In which structure do the pollen grains germinate?

A. 1
B. 2
C. 3
D. 4

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000078|!|

Which of the labelled structures will wither and fall off after fertilization?

A. 1 only
B. 2 and 3 only
C. 1, 2 and 3 only
D. 2, 3 and 4 only

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0130000079|!| ★

Directions: The following two questions refer to the diagram below which shows the flowers of a plant. (The parts are not drawn to the same scale.)



The flowers of this plant are

A. insect-pollinated because the petals are large and conspicuous.
B. insect-pollinated because the stigmas are feathery.
C. wind-pollinated because the ovary is outside the flower.
D. wind-pollinated because the anthers hang outside the flower.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000080|!| ★

The stigma of these flowers is adapted to

A. adhere the pollen grains from insects.
B. catch the pollen grains in the air.
C. attract insects for pollination.
D. facilitate the dispersal of pollen grains by wind.

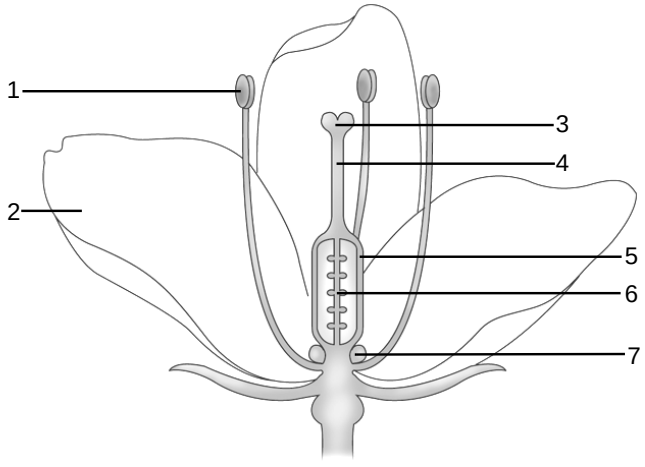
##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0130000081|!|

Directions: The following two questions refer to the following diagram which shows the structures of a flower.



Which of the following structures secretes sugary solutions to stimulate pollen grains of the same species to develop pollen tubes?

A. structure 1
B. structure 2
C. structure 3
D. structure 7

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000082|!|

Which of the following statements about the floral structures is/are correct?

(1) Structure 4 is a part of the carpel.
(2) Structure 5 will develop into a seed after fertilization.
(3) Structure 6 carries female gametes.

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

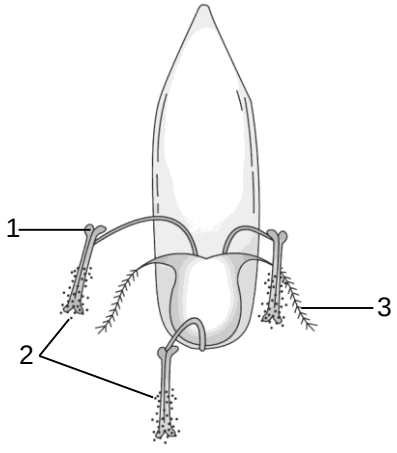
##B##

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Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000083|!|

Directions: The following two questions refer to the following diagram which shows a flower.



According to the diagram, which of the following statements about structures 2 is ***incorrect***?

A. They are light.
B. They can be transferred to the stigma of another flower by wind.
C. They are smooth.
D. They carry female gametes.

□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000084|!|

Which of the following statements account(s) for the fact that this is a wind-pollinated flower?

(1) Structure 1 hangs outside the flower.
(2) Structure 1 produces structure 2.
(3) Structure 3 is feathery.

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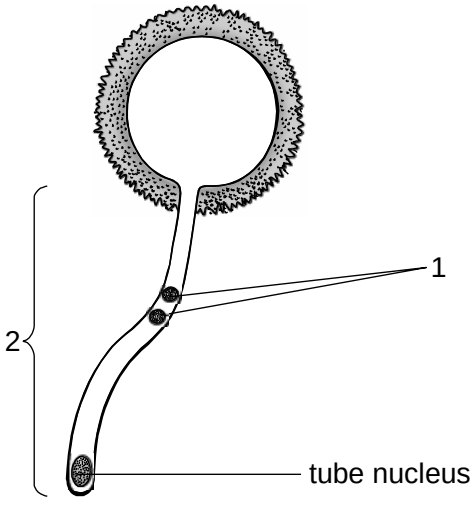
A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000085|!| ★

Directions: The following two questions refer to the following diagram which shows a germinating pollen grain.



Structures 1 are
A. vacuoles.
B. chloroplasts.
C. female gametes.
D. male gametes.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0130000086|!| ★

Which of the following statements about structure 2 is/are correct?

(1) It develops from a pollen grain in response to a sugary solution on the stigma.
(2) It is used to transfer the tube nucleus to the ovule for fertilization.
(3) It secretes enzymes to digest the tissues of the carpel for it to grow down.

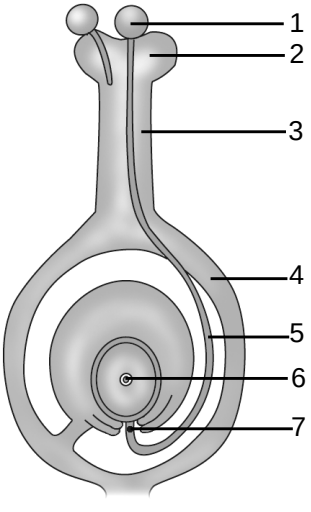
A. (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000087|!|

Directions: The following two questions refer to the following diagram which shows the process of fertilization in flowering plants.



Which of the following statements about structure 5 is/are correct?

(1) Structure 1 develops structure 5 in response to the sugary solution secreted by structure 2.
(2) Structure 5 secretes enzymes to digest the tissue of structure 3.
(3) The growth of structure 5 is towards the micropyle.

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A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0130000088|!|

Which of the following combinations about the fate of different structure after fertilization is correct?

	Structure	***After fertilization***
A.	3	becomes the fruit wall
B.	4	becomes the seed
C.	6	falls off
D.	7	becomes the embryo

##D##

Skill assessed	Mark
Analysing data	1